

2025년 1학기 데이터통신 10주차

과제 보고서

-TCP/UDP-



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과제 해결

TCP

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jellyfish319@jellyfish319-server:~$ iperf3 --server -p 5203
Server listening on 5203 (test #1)
Accepted connection from 192.168.42.130, port 51532
[ 5] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 51534
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-1.00 sec    668 MBytes  5.60 Gbits/sec
[ 5] 1.00-2.00 sec    716 MBytes  6.01 Gbits/sec
[ 5] 2.00-3.00 sec    750 MBytes  6.30 Gbits/sec
[ 5] 3.00-4.00 sec    752 MBytes  6.31 Gbits/sec
[ 5] 4.00-5.00 sec    743 MBytes  6.23 Gbits/sec
[ 5] 5.00-6.00 sec    745 MBytes  6.25 Gbits/sec
[ 5] 6.00-7.00 sec    746 MBytes  6.26 Gbits/sec
[ 5] 7.00-8.00 sec    744 MBytes  6.24 Gbits/sec
[ 5] 8.00-9.00 sec    764 MBytes  6.41 Gbits/sec
[ 5] 9.00-10.00 sec   759 MBytes  6.37 Gbits/sec
[ 5] 10.00-10.00 sec  2.12 MBytes  6.21 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-10.00 sec  7.22 GBytes  6.20 Gbits/sec
-----
Server listening on 5203 (test #2)
Accepted connection from 192.168.42.130, port 46534
[ 5] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 46546
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-1.00 sec    774 MBytes  6.48 Gbits/sec
[ 5] 1.00-2.00 sec    747 MBytes  6.27 Gbits/sec
[ 5] 2.00-3.00 sec    753 MBytes  6.32 Gbits/sec
[ 5] 3.00-4.00 sec    761 MBytes  6.39 Gbits/sec
[ 5] 4.00-5.00 sec    777 MBytes  6.52 Gbits/sec
[ 5] 5.00-6.00 sec    777 MBytes  6.52 Gbits/sec
[ 5] 6.00-7.00 sec    782 MBytes  6.55 Gbits/sec
[ 5] 7.00-8.00 sec    777 MBytes  6.52 Gbits/sec
[ 5] 8.00-9.00 sec    784 MBytes  6.58 Gbits/sec
[ 5] 9.00-10.00 sec   791 MBytes  6.64 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-10.00 sec  7.55 GBytes  6.48 Gbits/sec
-----
Server listening on 5203 (test #3)
-----
jellyfish319@jellyfish319-server:~$ iperf3 --client 192.168.42.129 -p 5202 -t 10
2
[1] * Stopped iperf3 --client 192.168.42.129 -p 5202 -t 10
jellyfish319@jellyfish319-server:~$ iperf3 --client 192.168.42.129 -p 5203 -t 10
Connecting to host 192.168.42.129, port 5203
[ 5] local 192.168.42.130 port 51534 connected to 192.168.42.129 port 5203
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 5] 0.00-1.00 sec    672 MBytes  5.62 Gbits/sec  101    1.72 MBytes
[ 5] 1.00-2.00 sec    712 MBytes  5.98 Gbits/sec   31    1.49 MBytes
[ 5] 2.00-3.00 sec    748 MBytes  6.27 Gbits/sec    0    1.81 MBytes
[ 5] 3.00-4.00 sec    759 MBytes  6.31 Gbits/sec   27    1.61 MBytes
[ 5] 4.00-5.00 sec    742 MBytes  6.23 Gbits/sec   40    1.38 MBytes
[ 5] 5.00-6.00 sec    746 MBytes  6.25 Gbits/sec    0    1.74 MBytes
[ 5] 6.00-7.00 sec    748 MBytes  6.27 Gbits/sec   38    1.53 MBytes
[ 5] 7.00-8.00 sec    742 MBytes  6.23 Gbits/sec    0    1.85 MBytes
[ 5] 8.00-9.00 sec    765 MBytes  6.42 Gbits/sec   38    1.67 MBytes
[ 5] 9.00-10.00 sec   759 MBytes  6.36 Gbits/sec   34    1.45 MBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 5] 0.00-10.00 sec  7.22 GBytes  6.20 Gbits/sec  309
[ 5] 0.00-10.00 sec  7.22 GBytes  6.20 Gbits/sec
-----
iperf Done.
jellyfish319@jellyfish319-server:~$ iperf3 --client 192.168.42.129 -p 5203 -t 10
Connecting to host 192.168.42.129, port 5203
[ 5] local 192.168.42.130 port 46546 connected to 192.168.42.129 port 5203
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 5] 0.00-1.00 sec    777 MBytes  6.51 Gbits/sec   72    1.73 MBytes
[ 5] 1.00-2.00 sec    745 MBytes  6.26 Gbits/sec   38    1.51 MBytes
[ 5] 2.00-3.00 sec    753 MBytes  6.31 Gbits/sec   25    1.30 MBytes
[ 5] 3.00-4.00 sec    762 MBytes  6.39 Gbits/sec    0    1.67 MBytes
[ 5] 4.00-5.00 sec    777 MBytes  6.52 Gbits/sec    7    1.45 MBytes
[ 5] 5.00-6.00 sec    777 MBytes  6.52 Gbits/sec    0    1.80 MBytes
[ 5] 6.00-7.00 sec    781 MBytes  6.55 Gbits/sec   21    1.62 MBytes
[ 5] 7.00-8.00 sec    777 MBytes  6.52 Gbits/sec   24    1.41 MBytes
[ 5] 8.00-9.00 sec    786 MBytes  6.59 Gbits/sec    0    1.67 MBytes
[ 5] 9.00-10.00 sec   795 MBytes  6.66 Gbits/sec    0    1.85 MBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 5] 0.00-10.00 sec  7.55 GBytes  6.48 Gbits/sec  188
[ 5] 0.00-10.00 sec  7.55 GBytes  6.48 Gbits/sec
-----
iperf Done.
jellyfish319@jellyfish319-server:~$
```

TCP의 경우 Transfer가 약 7.5GBytes, Bitrate가 6.3Gbits/sec 정도 나온다. Retr는 180~300회정도 발생하며 cwnd는 1.6Mbytes 정도 된다.

Retr는 재전송횟수로 손실된 데이터의 재전송을 통한 신뢰성이 보장되며 높은 데이터 전송량과 대역폭 안정성이 확보된다는 것이다. 또한, 네트워크 혼잡이 발생할 경우 cwnd가 조정되기에 네트워크의 현재 상황에 맞게 효율적으로 대응할 수 있다.

단, Retr 횟수가 증가하면 네트워크의 부하가 증가할 수 있기에 지연시간이 생길 우려가 있다.

UDP

<pre>Server listening on 5203 (test #3) ----- Accepted connection from 192.168.42.130, port 55898 [S] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 47562 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 1.13 MBytes 10.0 Mbits/sec 0.007 ms 0/863 (0%) [S] 1.00-2.00 sec 1.13 MBytes 10.0 Mbits/sec 0.011 ms 0/863 (0%) [S] 2.00-3.00 sec 1.13 MBytes 10.0 Mbits/sec 0.007 ms 0/863 (0%) [S] 3.00-4.00 sec 1.13 MBytes 10.0 Mbits/sec 0.012 ms 0/864 (0%) [S] 4.00-5.00 sec 1.13 MBytes 10.0 Mbits/sec 0.014 ms 0/863 (0%) [S] 5.00-6.00 sec 1.13 MBytes 10.0 Mbits/sec 0.010 ms 0/863 (0%) [S] 6.00-7.00 sec 1.13 MBytes 10.0 Mbits/sec 0.014 ms 0/864 (0%) [S] 7.00-8.00 sec 1.13 MBytes 9.99 Mbits/sec 0.011 ms 0/863 (0%) [S] 8.00-9.00 sec 1.13 MBytes 10.0 Mbits/sec 0.004 ms 0/864 (0%) [S] 9.00-10.00 sec 1.13 MBytes 9.99 Mbits/sec 0.006 ms 0/862 (0%) ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 11.9 MBytes 10.0 Mbits/sec 0.006 ms 0/8632 (0%) receiver Server listening on 5203 (test #4) ----- Accepted connection from 192.168.42.130, port 59594 [S] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 38855 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 11.9 MBytes 99.9 Mbits/sec 0.034 ms 0/8626 (0%) [S] 1.00-2.00 sec 11.9 MBytes 100 Mbits/sec 0.060 ms 0/8632 (0%) [S] 2.00-3.00 sec 11.9 MBytes 100 Mbits/sec 0.141 ms 0/8633 (0%) [S] 3.00-4.00 sec 11.9 MBytes 100 Mbits/sec 0.065 ms 0/8632 (0%) [S] 4.00-5.00 sec 11.9 MBytes 100 Mbits/sec 0.040 ms 0/8632 (0%) [S] 5.00-6.00 sec 11.9 MBytes 100 Mbits/sec 0.088 ms 0/8633 (0%) [S] 6.00-7.00 sec 11.9 MBytes 100 Mbits/sec 0.065 ms 0/8633 (0%) [S] 7.00-8.00 sec 11.9 MBytes 100 Mbits/sec 0.101 ms 0/8633 (0%) [S] 8.00-9.00 sec 11.9 MBytes 100 Mbits/sec 0.118 ms 0/8633 (0%) [S] 9.00-10.00 sec 11.9 MBytes 100 Mbits/sec 0.055 ms 0/8631 (0%) ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 119 MBytes 100 Mbits/sec 0.059 ms 0/86318 (0%) receiver Server listening on 5203 (test #5) -----</pre>	<pre>iperf3 --client 192.168.42.129 -p 5202 -t 10 -u -b 10M jellyfish319@jellyfish319-server:~\$ iperf3 --client 192.168.42.129 -p 5203 -t 10 -u -b 10M Connecting to host 192.168.42.129, port 5203 [S] local 192.168.42.130 port 47562 connected to 192.168.42.129 port 5203 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 1.00-2.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 2.00-3.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 3.00-4.00 sec 1.13 MBytes 10.0 Mbits/sec 864 [S] 4.00-5.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 5.00-6.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 6.00-7.00 sec 1.13 MBytes 10.0 Mbits/sec 864 [S] 7.00-8.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 8.00-9.00 sec 1.13 MBytes 10.0 Mbits/sec 863 [S] 9.00-10.00 sec 1.13 MBytes 9.99 Mbits/sec 863 ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 11.9 MBytes 10.0 Mbits/sec 0.000 ms 0/8632 (0%) sender [S] 0.00-10.00 sec 11.9 MBytes 10.0 Mbits/sec 0.006 ms 0/8632 (0%) receiver iperf Done. jellyfish319@jellyfish319-server:~\$ iperf3 --client 192.168.42.129 -p 5203 -t 10 -u -b 100M Connecting to host 192.168.42.129, port 5203 [S] local 192.168.42.130 port 38855 connected to 192.168.42.129 port 5203 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 11.9 MBytes 99.9 Mbits/sec 8626 [S] 1.00-2.00 sec 11.9 MBytes 100 Mbits/sec 8632 [S] 2.00-3.00 sec 11.9 MBytes 100 Mbits/sec 8633 [S] 3.00-4.00 sec 11.9 MBytes 100 Mbits/sec 8632 [S] 4.00-5.00 sec 11.9 MBytes 100 Mbits/sec 8633 [S] 5.00-6.00 sec 11.9 MBytes 100 Mbits/sec 8633 [S] 6.00-7.00 sec 11.9 MBytes 100 Mbits/sec 8632 [S] 7.00-8.00 sec 11.9 MBytes 100 Mbits/sec 8633 [S] 8.00-9.00 sec 11.9 MBytes 100 Mbits/sec 8633 [S] 9.00-10.00 sec 11.9 MBytes 100 Mbits/sec 8632 ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 119 MBytes 100 Mbits/sec 0.000 ms 0/86319 (0%) sender [S] 0.00-10.00 sec 119 MBytes 100 Mbits/sec 0.059 ms 0/86318 (0%) receiver iperf Done. jellyfish319@jellyfish319-server:~\$</pre>
<pre>Server listening on 5203 (test #6) ----- Accepted connection from 192.168.42.130, port 54268 [S] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 48941 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 63.8 MBytes 535 Mbits/sec 0.017 ms 3039/49217 (6.2%) [S] 1.00-2.00 sec 62.9 MBytes 527 Mbits/sec 0.021 ms 3520/49041 (7.2%) [S] 2.00-3.00 sec 62.4 MBytes 523 Mbits/sec 0.025 ms 3936/49117 (8%) [S] 3.00-4.00 sec 60.6 MBytes 508 Mbits/sec 0.018 ms 5301/49677 (12%) [S] 4.00-5.00 sec 57.7 MBytes 484 Mbits/sec 0.019 ms 7549/49368 (15%) [S] 5.00-6.00 sec 61.7 MBytes 518 Mbits/sec 0.030 ms 4717/49407 (9.5%) [S] 6.00-7.00 sec 59.5 MBytes 499 Mbits/sec 0.024 ms 6508/49581 (13%) [S] 7.00-8.00 sec 60.6 MBytes 508 Mbits/sec 0.023 ms 5351/49231 (11%) [S] 8.00-9.00 sec 56.4 MBytes 473 Mbits/sec 0.040 ms 8059/49663 (16%) [S] 9.00-10.00 sec 59.6 MBytes 475 Mbits/sec 0.036 ms 8485/49595 (17%) [S] 10.00-10.00 sec 126 KBytes 705 Mbits/sec 0.049 ms 51/140 (36%) ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 602 MBytes 505 Mbits/sec 0.049 ms 57807/493947 (12%) receiver Server listening on 5203 (test #6) ----- Accepted connection from 192.168.42.130, port 54184 [S] local 192.168.42.129 port 5203 connected to 192.168.42.130 port 54918 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 61.0 MBytes 512 Mbits/sec 0.020 ms 4624/48783 (9.5%) [S] 1.00-2.00 sec 60.2 MBytes 505 Mbits/sec 0.021 ms 6380/50005 (13%) [S] 2.00-3.00 sec 55.2 MBytes 463 Mbits/sec 0.027 ms 9124/49115 (19%) [S] 3.00-4.00 sec 61.0 MBytes 512 Mbits/sec 0.020 ms 5594/49745 (11%) [S] 4.00-5.00 sec 60.4 MBytes 507 Mbits/sec 0.027 ms 5815/49558 (12%) [S] 5.00-6.00 sec 58.6 MBytes 492 Mbits/sec 0.025 ms 7099/49543 (14%) [S] 6.00-7.00 sec 60.1 MBytes 504 Mbits/sec 0.047 ms 5646/49136 (11%) [S] 7.00-8.00 sec 57.8 MBytes 485 Mbits/sec 0.027 ms 7611/49465 (15%) [S] 8.00-9.00 sec 56.8 MBytes 476 Mbits/sec 0.022 ms 8812/49329 (18%) [S] 9.00-10.00 sec 57.0 MBytes 479 Mbits/sec 0.030 ms 8040/49350 (16%) [S] 10.00-10.00 sec 102 KBytes 490 Mbits/sec 0.023 ms 28/157 (18%) ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 588 MBytes 493 Mbits/sec 0.023 ms 68763/494784 (14%) receiver Server listening on 5203 (test #7) -----</pre>	<pre>iperf Done. jellyfish319@jellyfish319-server:~\$ iperf3 --client 192.168.42.129 -p 5203 -t 10 -u -b 10M Connecting to host 192.168.42.129, port 5203 [S] local 192.168.42.130 port 48941 connected to 192.168.42.129 port 5203 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 68.0 MBytes 570 Mbits/sec 49237 [S] 1.00-2.00 sec 67.8 MBytes 569 Mbits/sec 49113 [S] 2.00-3.00 sec 67.9 MBytes 569 Mbits/sec 49149 [S] 3.00-4.00 sec 68.6 MBytes 575 Mbits/sec 49571 [S] 4.00-5.00 sec 68.1 MBytes 572 Mbits/sec 49446 [S] 5.00-6.00 sec 68.2 MBytes 572 Mbits/sec 49421 [S] 6.00-7.00 sec 68.4 MBytes 574 Mbits/sec 49538 [S] 7.00-8.00 sec 68.1 MBytes 574 Mbits/sec 49285 [S] 8.00-9.00 sec 68.6 MBytes 576 Mbits/sec 49597 [S] 9.00-10.00 sec 68.3 MBytes 573 Mbits/sec 49430 ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 682 MBytes 572 Mbits/sec 0.000 ms 0/493947 (0%) sender [S] 0.00-10.00 sec 682 MBytes 505 Mbits/sec 0.049 ms 57807/493947 (12%) receiver iperf Done. jellyfish319@jellyfish319-server:~\$ iperf3 --client 192.168.42.129 -p 5203 -t 10 -u -b 2G Connecting to host 192.168.42.129, port 5203 [S] local 192.168.42.130 port 54918 connected to 192.168.42.129 port 5203 [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-1.00 sec 67.5 MBytes 566 Mbits/sec 48094 [S] 1.00-2.00 sec 69.0 MBytes 579 Mbits/sec 49974 [S] 2.00-3.00 sec 67.9 MBytes 570 Mbits/sec 49187 [S] 3.00-4.00 sec 68.7 MBytes 576 Mbits/sec 49753 [S] 4.00-5.00 sec 68.3 MBytes 573 Mbits/sec 49461 [S] 5.00-6.00 sec 68.5 MBytes 575 Mbits/sec 49597 [S] 6.00-7.00 sec 67.9 MBytes 570 Mbits/sec 49181 [S] 7.00-8.00 sec 68.3 MBytes 573 Mbits/sec 49449 [S] 8.00-9.00 sec 69.0 MBytes 578 Mbits/sec 49334 [S] 9.00-10.00 sec 68.2 MBytes 572 Mbits/sec 49354 ----- [ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams [S] 0.00-10.00 sec 683 MBytes 573 Mbits/sec 0.000 ms 0/494784 (0%) sender [S] 0.00-10.00 sec 588 MBytes 493 Mbits/sec 0.023 ms 68763/494784 (14%) receiver iperf Done. jellyfish319@jellyfish319-server:~\$</pre>

UDP의 경우 bandwidth 옵션에 따라 10M, 100M에서는 손실률이 0%이며 1G, 2G일 때는 14%의 손실이 나온다. 또한, Bitrate의 경우 1G, 2G에서는 493Mbits/sec로 제한됨을 확인할 수 있다.

네트워크 여유가 충분한 조건 (10M, 100M) 하에서는 데이터 손실 없이 빠른 전송이 가능하며 TCP 대비 재전송과 같은 과정이 없으므로 시스템 부하가 적어 오버헤드가 적다는 장점이 있다. 따라서 latency가 낮으며 실시간 전송에 유리함을 파악할 수 있다.

그러나, 1G, 2G처럼 과도한 전송 속도를 요구할 경우 패킷손실률이 증가하며 UDP는 자동 재전송 기능이 없기에 데이터에 대한 신뢰성이 떨어질 수 있다. 또한, 설정한 대역폭에 비해 실제 대역폭이 낮게 나온다는 단점이 있다.